CHAPTER D4

Evaluation, testing, and optimization

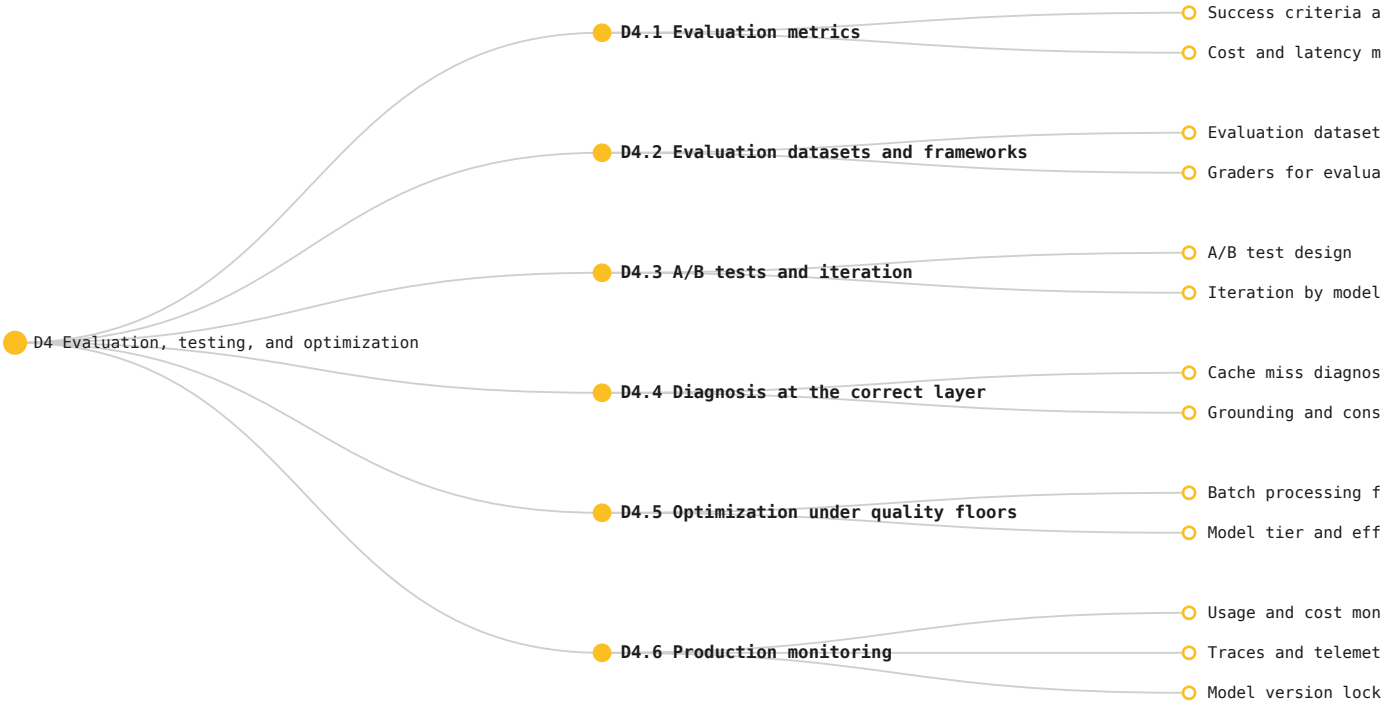
WEIGHT 16%. INFERENCE FROM SECONDARY SOURCES.

Measure outcomes, diagnose the layer, batch what can wait.

6 sections, 13 components, 18 recall cues.

CONCEPT MAP

Units branch to components. On a small screen, scroll sideways.



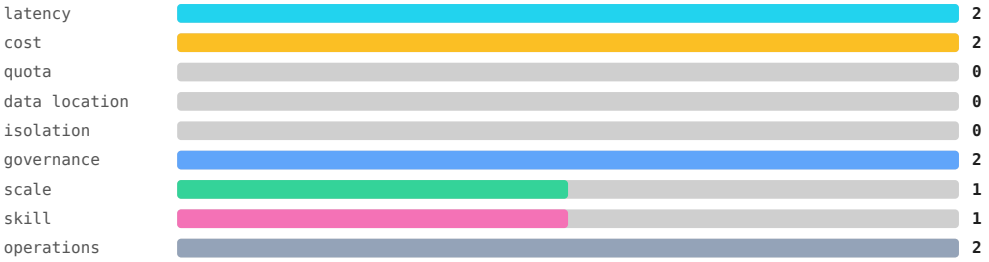
SECTIONS

D4.1 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

Evaluation metrics

Business outcome is primary. Proxy scores are diagnostics.

CONSTRAINT FAMILIES



SUCCESS CRITERIA AND PRIMARY METRIC

A metric measures the goal only when it names the goal and a threshold before tuning starts.

Rule Make the business outcome the primary metric, and keep proxy scores as diagnostics.

Trap A fluency or length score that rises while task success stays flat. The score looks better, but it does not measure the goal.

Exam tip A metric with no threshold and no owner cannot decide a release.

COST AND LATENCY METRICS

You pay for completed tasks, so cost per completed task measures value better than cost per token.

Rule Report spend next to completed tasks and the quality threshold in the same view.

Trap A lower price per token after a change. Fewer tasks may complete, so the cost per completed task can rise.

Exam tip A cost figure without a completed-task count does not answer a cost question.

▼ RECALL CUES (3)

1. **a proxy score rises while the business outcome stays flat**
   Answer primary metric on the business outcome
   Rule Make the outcome the primary metric with a threshold. Keep the proxy as a diagnostic.
   - Tune the proxy harder
   - Move to a larger model tier before the measurement
2. **the team needs a number for a vague quality goal**
   Answer criterion with a threshold and an owner
   Rule Turn the goal into a measurable criterion, then attach a threshold and an owner.
   - Report a qualitative impression
   - Add more logging with no threshold
3. **spend rises after a change that adds more tasks to the queue**
   Answer cost per completed task
   Rule Divide spend by completed tasks and compare that value across versions.
   - Compare cost per token only
   - Cut tokens without a check of the quality threshold

▼ NUMBERS AND VERSIONS (4)

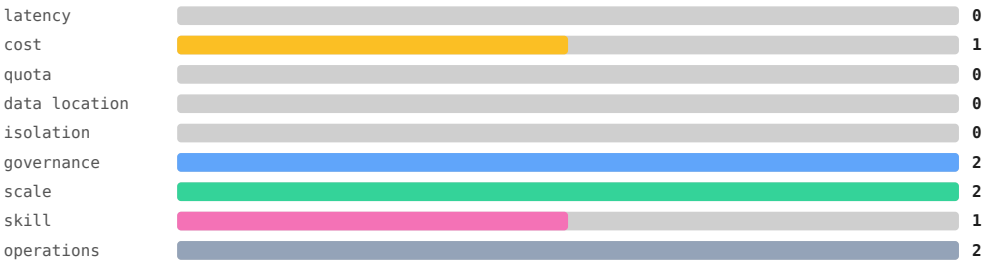
- The docs example sets an F1 score of at least 0.85 on a held-out test set of 10,000 posts. source
- The Usage API supports 1m, 1h, and 1d time buckets. source
- The default limit for 1m buckets is 60, and the maximum is 1,440 buckets. source
- Usage and cost data typically appears within 5 minutes of request completion. source

D4.2 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

Evaluation datasets and frameworks

Build from production traffic. Add production edge cases.

CONSTRAINT FAMILIES



EVALUATION DATASET DESIGN

An eval set measures the system well only when its cases match the real task distribution.

Rule Build the eval set from production traffic, and add edge cases that production contains.

Trap A set built only from easy, clean examples. The score is high, but it does not predict production behavior.

Exam tip A high average on a clean set does not prove the system handles the hard cases.

GRADERS FOR EVALUATIONS

Use the cheapest grader that stays reliable for each type of case.

Rule Assign each case type the cheapest grader that passed a reliability test on that type.

Trap An LLM grader that runs on every case with no reliability test. The cost is low, but the scores may be wrong.

Exam tip Human graders are the costliest rung, so they fit only the cases that code and LLM graders cannot decide.

▼ RECALL CUES (3)

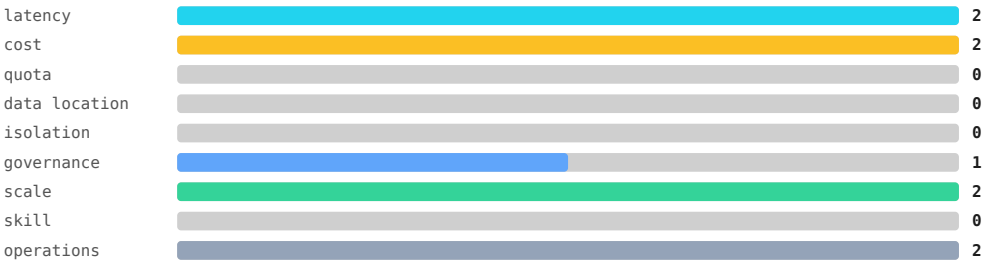
1. **a high average score on a clean set while production fails**
   Answer edge and adversarial cases added to the set
   Rule Add the cases that production contains. Report scores per case type.
   - Raise the clean-set average further
   - Add a larger model tier
2. **grader cost is too high for a daily run**
   Answer cheapest reliable grader per case type
   Rule Use code first. Use an LLM grader only after a reliability test.
   - Grade every case by hand
   - Skip the reliability test to save time
3. **an LLM grader gives scores that people do not trust**
   Answer rubric and reliability test
   Rule Write a rubric with a fixed output format. Test the grader before it scores releases.
   - Accept the grader score without a test
   - Drop the grader and keep no check

D4.3 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

A/B tests and iteration

One arm per unit. Compare on the primary metric and threshold.

CONSTRAINT FAMILIES



A/B TEST DESIGN

A control arm shows whether the change, and not the traffic mix, caused the measured difference.

Rule Assign each unit to one arm, and compare the arms on the primary metric with a stated threshold.

Trap A full rollout that compares the week before with the week after. The traffic mix changed, so the difference does not show the effect of the change.

Exam tip A significant result that fails the business value check does not justify the switch. (Inference)

ITERATION BY MODEL AND EFFORT

A newer model or higher effort earns its extra cost only when it completes more tasks.

Rule Ship a model or effort change only when it clears the quality floor and lowers or holds cost per completed task.

Trap A switch to a larger model because its score is higher. The score rises, but the cost per completed task may rise faster than the gain.

Exam tip A higher per-token price does not decide the switch. Compare the completed-task cost instead.

▼ RECALL CUES (3)

1. **a full rollout shows a small score move with no control**
   Answer A/B test with a control arm
   Rule Assign units to a control and a candidate arm, and compare one primary metric.
   - Compare two time periods
   - Ship and watch the next week
2. **the test shows a real lift that costs more than it returns**
   Answer business significance check
   Rule Separate statistical significance from business value. Decide with the cost of the switch.
   - Ship because the lift is significant
   - Run the test longer to find a bigger lift
3. **a candidate change touches tools or permissions**
   Answer shadow run before exposure
   Rule Measure the candidate on logged or copied traffic before users see it. (Inference)
   - Expose all users at once
   - Skip the guard metrics

▼ NUMBERS AND VERSIONS (2)

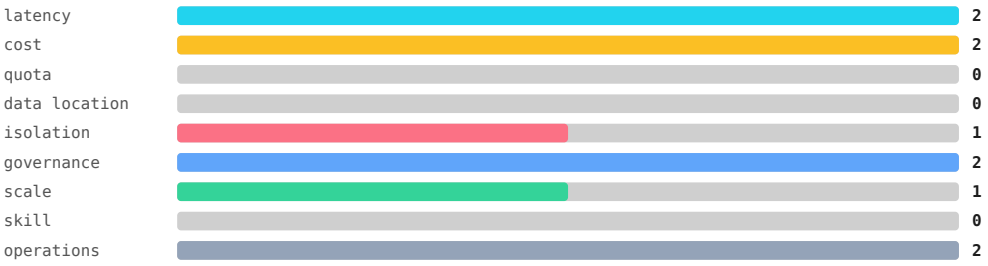
- In Anthropic tests, low effort with failures re-run at high cost about 40% less than all tasks at high, at the same pass rate. source
- Claude Opus 5.5 and Claude Haiku 5.5 default to medium effort. Other models that support effort default to high. source

D4.4 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

Diagnosis at the correct layer

Earliest changed block in the prefix. Move variables after it.

CONSTRAINT FAMILIES



CACHE MISS DIAGNOSIS

The earliest divergence in a request explains the miss, so find that divergence before any other fix.

Rule Find the earliest changed block in the prefix, and move the variable content after the breakpoint.

Trap A bigger model tier to recover a lower hit rate. The tier change adds cost, and the prefix still changes on every request.

Exam tip Zero cache read and zero cache write counts point to a prompt below the minimum cacheable length, not to a model choice.

GROUNDING AND CONSISTENCY CHECKS

An unsupported claim is easier to catch when the answer must point to its source text.

Rule Require a quote or citation for each claim, and allow the model to say that it does not know.

Trap A stronger model added to fix invented facts. The model can still invent facts without a check that ties each claim to a source.

Exam tip Answers that change across runs for the same input signal a possible hallucination that a check can catch.

▼ RECALL CUES (3)

1. **an answer states a figure that no source supports**
   Answer grounding check with quotes and citations
   Rule Ask for quotes first, then require a citation for each claim.
   - Use a larger model tier
   - Add tools without a grounding step
2. **quality drops after a prompt edit while the model stays the same**
   Answer layer-by-layer diagnosis with a split test
   Rule Name the changed layer first, then test the suspects one at a time.
   - Upgrade the model tier
   - Edit several layers at once
3. **cache reads fall to zero after a template edit**
   Answer earliest divergence fix
   Rule Read cache_miss_reason, fix the earliest divergence, and confirm with cache read counts.
   - Raise the cache TTL
   - Add breakpoints on the changed block

▼ NUMBERS AND VERSIONS (1)

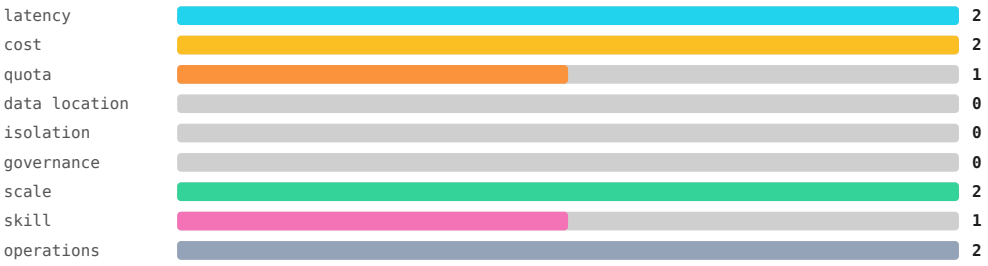
- The docs say to ask for word-for-word quotes first on documents longer than 20k tokens. source

D4.5 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

Optimization under quality floors

Work that nobody waits for: batch. User waits: synchronous.

CONSTRAINT FAMILIES



BATCH PROCESSING FOR WORK THAT CAN WAIT

Work that no user waits for can trade latency for a lower token price.

Rule Send unattended work to the batch interface, and keep any request a user waits for synchronous.

Trap Streaming a nightly job to cut its latency. Streaming changes the timing of the first token, and the job still pays the synchronous price.

Exam tip A job with a deadline of minutes does not fit batch processing.

MODEL TIER AND EFFORT CHOICE

The cheapest model and effort that clear the quality floor give the lowest cost per completed task.

Rule Choose the cheapest model and effort that clear the quality floor on each task route.

Trap Routing all traffic to the cheapest tier. The cost falls, but hard tasks fail the quality floor.

Exam tip A speed gain is valid only when the quality floor still holds after the change.

▼ RECALL CUES (3)

1. **a nightly job must score the eval set and no user waits**
   Answer batch processing
   Rule Send unattended work to the batch interface. Keep calls that users see synchronous.
   - Stream the nightly job
   - Move the job to a faster tier with no deadline
2. **latency is above the target and the quality floor is close**
   Answer measured cut under floors
   Rule Measure each stage, then cut the lever with the least quality cost first. Check the floor after each cut.
   - Cut context and caching together with no measure
   - Move all traffic to the cheapest tier
3. **some tasks are easy enough for a cheaper tier**
   Answer routing with a floor per route
   Rule Route only the task types that the cheaper tier handles above the floor.
   - Route everything to the cheapest tier
   - Route by guess with no measurement

▼ NUMBERS AND VERSIONS (9)

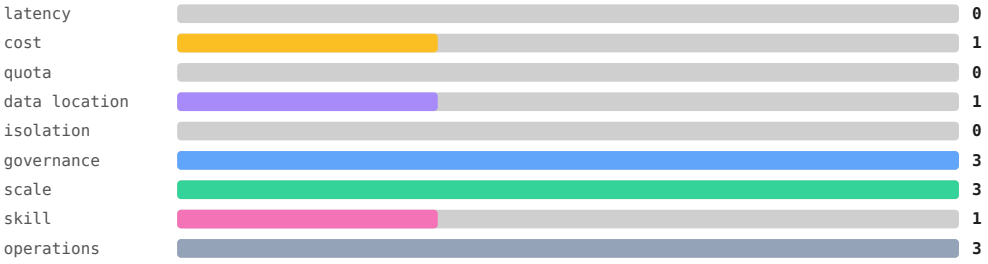
- The Batch API takes 50 percent off every token of a request, cached tokens too. source
- A batch holds up to 100,000 Message requests or 256 MB, whichever limit comes first. source
- Most batches complete within 1 hour. source
- Results become available after all messages complete or after 24 hours, whichever comes first. source
- Batch results are available for 29 days after creation. source
- A batched request must set max_tokens to at least 1. source
- The docs say Claude Haiku 5.5 offers the fastest response times with high intelligence. source
- The docs call effort the main control for speed and cost on Claude Haiku 5.5. source
- The docs say to start most agent workloads with Claude Opus 5.5 at default effort medium. source

D4.6 CHAPTER D4, EVALUATION, TESTING, AND OPTIMIZATION

Production monitoring

Show spend beside completed tasks. Alert on cost per task.

CONSTRAINT FAMILIES



USAGE AND COST MONITORING

Spend becomes a quality signal only when it is divided by completed tasks in the same view.

Rule Show spend next to completed tasks, and alert on the per-task line.

Trap A dashboard of total tokens with no completed-task count. Total spend looks normal while the outcome rate falls.

Exam tip A cost alert with no owner is a notification, not a control. (Inference)

TRACES AND TELEMETRY

A trace shows the order and timing of each hop in one request, which logs alone do not show.

Rule Add a destination per data rule, and choose the content mode that the rule allows.

Trap Application logs alone. They record events, but they do not show the order and timing of each hop.

Exam tip Full content gives the most detail, and it stores prompts and responses in the destination.

MODEL VERSION LOCK AND DRIFT

A pinned model ID fixes the weights, so a behavior shift points to infrastructure, prompts, or data.

Rule Pin the model ID, keep a baseline, and compare live metrics with that baseline after each change.

Trap A belief that a model ID guarantees identical output at all times. The weights stay fixed, but infrastructure updates can shift observable behavior.

Exam tip A stable model ID does not rule out a behavior shift from infrastructure.

▼ RECALL CUES (3)

1. **usage totals rise while completed tasks stay flat**
   Answer cost per completed task line with an owner
   Rule Show spend next to completed tasks, and alert on the per-task line with a named owner.
   - Alert on total spend only
   - Cut the model tier before the data check
2. **live quality slides while the offline eval set still scores well**
   Answer alert with a refreshed production sample
   Rule Page the named owner, then refresh the eval set from production samples.
   - Trust the offline score
   - Roll back the model tier before diagnosis
3. **behavior shifts on a stable model ID with no code change**
   Answer baseline comparison and layer check
   Rule Compare live metrics with the baseline, then check prompts, data, and the model ID.
   - Assume a model ID fixes all output
   - Change the prompt before the evidence

▼ NUMBERS AND VERSIONS (7)

- The Usage API supports 1m, 1h, and 1d time buckets. source
- The default limit for 1h buckets is 24, and the maximum is 168 buckets. source
- Usage and cost data typically appears within 5 minutes of request completion. source
- With Full content, each span holds up to 0.75 MB of content by default. source
- HIPAA, FERPA, or K-12 restrictions make destinations export only metadata. source
- Claude Code exports metrics every 60000 ms by default. source
- For the 4.6 generation and later, the dateless model ID maps to a single fixed snapshot. source